

```
Last login: Mon Jun 22 11:33:30 on ttys001
adityadave@Adityas-MacBook-Air-3 ~ % cd /Users/adityadave/PE/PE-Daywise-Work-Tracker/19-JUNE/Q2\&Q21
adityadave@Adityas-MacBook-Air-3 Q2&Q21 % python3 script.py
Running 1/10...
Running 2/10...
Running 3/10...
Running 4/10...
Running 5/10...
Running 6/10...
Running 7/10...
Running 8/10...
Running 9/10...
Running 10/10...
```

=====

10TH RUN OUTPUT

=====

SET

WARNING: Log 1: Planner Input:

DETAIL: {QUERY

```
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname part
        :colnames ("p_partkey" "p_name" "p_mfg" "p_brand" "p_type" "p_size"
          "p_container" "p_retailprice" "p_comment")
      }
    :rtekind 0
    :relid 17169
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname supplier
        :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
          s_acctbal" "s_comment")
      }
    :rtekind 0
    :relid 17181
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals (
      {BOOLEXP
        :bbool and
        :args (
          {OPEXP
            :opno 1756
            :opfuncid 1720
            :opresulttype 16
            :opretset false
            :opcollid 0
            :inputcollid 0
          }
          :args (
            {VAR
              :varno 2
              :varattno 6
              :vartype 1700
              :vartypmod 983046
              :varcollid 0
              :varnullingrels (b)
              :varlevelsup 0
            }
            :varreturningtype 0
            :varnosyn 2
            :varattnosyn 6
            :location -1
          )
          {FUNCEXP
            :funcid 1740
            :funcresulttype 1700
            :funcresultset false
            :funcvariadic false
            :funcformat 2
            :funccollid 0
            :inputcollid 0
          }
          :args (
            {CONST
              :consttype 23
              :consttypmod -1
              :constcollid 0
              :constlen 4
              :constbyval true
              :constisnull false
              :location -1
              :constvalue 4 [ 0 0 0 0 0 0 0 ]
            }
          )
        )
      }
    :location -1
  )
}
:location -1
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:opetName <>
:subselect
  {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSublinks false
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity false
    :hasGroupRTE false
    :isReturn false
    :ctelist <>
```

```

:rtable (
  {RANGETBLENTY
  :alias
    {ALIAS
      :aliasname n
      :colnames <>
    }
  :eref
    {ALIAS
      :aliasname n
      :colnames ("n_nationkey" "n_name" "n_regionkey" "n_commen
t")
    }
  :rtekind 0
  :relid 17158
  :inh true
  :relkind r
  :rellockmode 1
  :perminfoindex 1
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals <>
  }
  {RANGETBLENTY
  :alias
    {ALIAS
      :aliasname r
      :colnames <>
    }
  :eref
    {ALIAS
      :aliasname r
      :colnames ("r_regionkey" "r_name" "r_comment")
    }
  :rtekind 0
  :relid 17164
  :inh true
  :relkind r
  :rellockmode 1
  :perminfoindex 2
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
  :relid 17158
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 8 10)
  :insertedCols (b)
  :updatedCols (b)
  }
  {RTEPERMISSIONINFO
  :relid 17164
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 8 9)
  :insertedCols (b)
  :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
    {RANGETBLREF
      :rtindex 2
    }
  )
  :quals
    {BOOLEXP
    :boolop and
    :args (
      {OPEXP
      :opno 96
      :opfncid 65
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location -1
        }
        {VAR
        :varno 2
        :varattno 4
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 1
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 4
        :location -1
        }
      )
    :location -1
    }
  )
  {OPEXP
  :opno 96
  :opfncid 65
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
    }
    {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
  }
  )
  }
  }

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        :varnosyn 2
        :varattnosyn 1
        :location -1
    }
}
:location -1
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
    {VAR
        :varno 2
        :varattno 2
        :vartype 1042
        :vartypmod 29
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 2
        :location -1
    }
    {CONST
        :consttype 1042
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
    }
)
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull false
            :location -1
            :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
        }
        :resno 1
        :resname ?column?
        :resortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}
}
{RANGETBENTRY
:alias <>
:eref
    {ALIAS
        :aliasname partsupp
        :colnames ("ps_partkey" "ps_supkey" "ps_availqty" "ps_supplycost" "p
            s_comment")
    }
:rtekind 0
:relid 17191
:inh true
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
    {ALIAS
        :aliasname nation
        :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
    }
:rtekind 0
:relid 17158
:inh true
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
    {ALIAS
        :aliasname region
        :colnames ("r_regionkey" "r_name" "r_comment")
    }
:rtekind 0
:relid 17164
:inh true
:relkind r
:rellockmode 1
:perminfoindex 5

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:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17169
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17181
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17191
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
{RANGETBLREF
:rtindex 5
}
)
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 283
}
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 295
}
)
:location 293
}
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 318
}
{VAR
:varno 3
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location 330
}
)
)
}
)

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```

:location 328
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 353
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 362
:constvalue 4 [ 15 0 0 0 0 0 0 ]
}
)
:location 368
}
{OPEXPR
:opno 1209
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{RELABELTYPE
:arg
{VAR
:varno 1
:varattno 5
:vartype 1043
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 377
}
:resulttype 25
:resulttypmod -1
:resultcollid 100
:relabelformat 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 389
:constvalue 10 [ 40 0 0 0 37 66 82 65 83 83 ]
}
)
:location 384
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location 410
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 424
}
)
:location 422
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location 448
}
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location 462
}
}

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    }
}
:location 460
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
:varno 5
:varattno 2
:varstype 1042
:varstypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 2
:location 493
}
  {CONST
:consttype 1042
:conststypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 509
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location 507
}
{OPEXPR
:opno 1752
:opfuncid 1718
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 3
:varattno 4
:varstype 1700
:varstypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 4
:location 530
}
  {SUBLINK
:sublinkType 4
:sublinkId 0
:testsExpr <>
:operName <>
:subselect
  {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
:alias <>
:eref
  {ALIAS
:aliasname partsupp
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "ps_comment")
}
:rtekind 0
:relid 17191
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
  {RANGETBENTRY
:alias <>
:eref
  {ALIAS
:aliasname supplier
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 17181
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
  {BOOLEXP
:boolop and
:args (
  {OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 2
:varattno 6
:varstype 1700
:varstypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location -1
}
  {FUNCEXP
:funcid 1740
:funcresulttype 1700
}
}
}
}
}
}

```

```

:funcreset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
  {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 0 0 0 0 0 0 0 0 ]
  }
)
:location -1
}
}
:location -1
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
  {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSubLinks false
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity false
    :hasGroupRTE false
    :isReturn false
    :ctelist <>
    :rtable (
      {RANGETBENTRY
        :alias
        {ALIAS
          :aliasname n
          :colnames <>
        }
        :eref
        {ALIAS
          :aliasname n
          :colnames ("n_nationkey" "n_name" "n_regio
            nkey" "n_comment")
        }
        :rtekind 0
        :relid 17158
        :inh true
        :relkind r
        :rellockmode 1
        :perminfoindex 1
        :tablesample <>
        :lateral false
        :inFromCl true
        :securityQuals <>
      }
      {RANGETBENTRY
        :alias
        {ALIAS
          :aliasname r
          :colnames <>
        }
        :eref
        {ALIAS
          :aliasname r
          :colnames ("r_regionkey" "r_name" "r_comme
            nt")
        }
        :rtekind 0
        :relid 17164
        :inh true
        :relkind r
        :rellockmode 1
        :perminfoindex 2
        :tablesample <>
        :lateral false
        :inFromCl true
        :securityQuals <>
      }
    )
    :rteperminfos (
      {RTEPERMISSIONINFO
        :relid 17158
        :inh true
        :requiredPerms 2
        :checkAsUser 0
        :selectedCols (b 8 10)
        :insertedCols (b)
        :updatedCols (b)
      }
      {RTEPERMISSIONINFO
        :relid 17164
        :inh true
        :requiredPerms 2
        :checkAsUser 0
        :selectedCols (b 8 9)
        :insertedCols (b)
        :updatedCols (b)
      }
    )
  )
  :jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
      {RANGETBLREF
        :rtindex 2
      }
    )
  }
  :quals
  {BOOLEXP
    :boolop and
    :args (
      {OPEXPR
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location -1
          }
        )
      }
    )
  }

```

```

    }
    {VAR
      :varno 2
      :varattno 4
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 1
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 4
      :location -1
    }
  }
  :location -1
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
  }
  {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
  }
)
:location -1
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
    :varno 2
    :varattno 2
    :vartype 1042
    :vartypmod 29
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 2
    :location -1
  }
  {CONST
    :consttype 1042
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 10 [ 40 0 0 0 69 85 82 7
      9 80 69 ]
  }
)
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location -1
      :constvalue 4 [ 1 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}
{RANGETBLENTY
:alias <>

```

```

:eref
{
  ALIAS
  :aliasname nation
  :colnames ("n_nationkey" "n_name" "n_regionkey" "n_com
ment")
}
:rtekind 0
:relid 17158
:inh true
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{
  ALIAS
  :aliasname region
  :colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 17164
:inh true
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17191
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17181
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 812
}
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 824
}
}
:location 822
}
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0

```

```

        :varnosyn 2
        :varattnosyn 1
        :location 863
    }
    {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location 875
    }
}
:location 873
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location 914
}
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 928
}
)
:location 926
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 3
:location 968
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 982
}
)
:location 980
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 4
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 1029
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 1045
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location 1043
}
}
:location 859
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2146
:aggtype 1700
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes (o 1700)
:aggdirectargs <>
:args (
{TARGETENTRY

```

```

:expr
{VAR
:varno 1
:varattno 4
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 599
}
:resno 1
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar false
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggno -1
:aggtransno -1
:location 595
}
:resno 1
:resname min
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 546
}
)
:location 544
}
)
:location 314
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location 51
}
:resno 1
:resname s_acctbal
:ressortgrouppref 1
:resorigtbl 17181
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location 70
}
:resno 2
:resname s_name
:ressortgrouppref 3
:resorigtbl 17181
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 86
}
:resno 3
:resname n_name
:ressortgrouppref 2
:resorigtbl 17158
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1

```

```

:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 102
}
:resno 4
:resname p_partkey
:ressortgroupref 4
:resorigtbl 17169
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 121
}
:resno 5
:resname p_mfgr
:ressortgroupref 0
:resorigtbl 17169
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 3
:vartype 1043
:vartypmod 44
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location 137
}
:resno 6
:resname s_address
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 5
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 5
:location 156
}
:resno 7
:resname s_phone
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 7
:vartype 1043
:vartypmod 105
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 7
:location 173
}
:resno 8
:resname s_comment
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 7
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 1752
:sortop 1756
:reverse_sort true
:nulls_first true
:hashable true
}
{SORTGROUPCLAUSE
:tleSortGroupRef 2
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
{SORTGROUPCLAUSE
:tleSortGroupRef 3
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
{SORTGROUPCLAUSE
:tleSortGroupRef 4
:eqop 96
:sortop 97
:reverse_sort false

```

```

:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount
{FUNCEXP
:funcid 481
:funcresulttype 20
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 1153
:constvalue 4 [ 100 0 0 0 0 0 0 0 ]
}
)
:location -1
}
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=481 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtableindex 1
}
{RANGETBLREF
:rtableindex 2
}
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23

```

```

:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 1
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
}
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
}
:location -1
}
}
{OEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
}
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

```

WARNING: is_parallel_safe: INPUT qual = ({RESTRICINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: max_parallel_hazard_walker: Param paramid=1 in safe_list=0 > reason=outer reference

WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPAGATE_RESTRICTED

WARNING: is_parallel_safe: INPUT qual = ({RESTRICINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [40 0 0 0 69 85 82 79 80 69]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b 2) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [40 0 0 0 69 85 82 79 80 69]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype

```

1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :c
onstbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false
:leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause
<> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042
:vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_
jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :con
stvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :le
ft_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:reldid 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:reldid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:reldid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:reldid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtableindex 1
}
{RANGETBLREF
:rtableindex 2
}
)
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
}
:location -1
}
{OPEXPR

```

```

:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
    :varno 2
    :varattno 2
    :vartype 1042
    :vartypmod 29
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 2
    :location -1
  }
  {CONST
    :consttype 1042
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
  }
}
:location -1
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :groupbustingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname <>
    :resortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:conflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupingSets false
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
{VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
{VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1}) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype
1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :co
nstbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :co
nstbyval false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 2) :right_rels (b) :orclause <>
:rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selc 0 :outer_selc -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042
:varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_j
domain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :cons
tvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :lef
t_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype
1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :c
onstbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false
:leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 2) :right_rels (b) :orclause
<> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selc -1 :outer_selc -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042
:varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_j
domain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :con
stvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :lef
t_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :va
rtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (
b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}) :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_rels (b 1 2) :required_rels (b 1 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b 2) :orclause <> :rinfo_s
erial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selc 0.0058823529411764705 :outer_selc -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :varattno 3 :vartype
2 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jd
omain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varn
osyn 2 :varattnosyn 1 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bu
cketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels
(b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_rels (b 1 2) :required_rels (b 1 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b 2) :orclause <> :rinfo_s
erial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selc 0.0058823529411764705 :outer_selc -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype
23 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_j
domain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varn
osyn 2 :varattnosyn 1 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_b
ucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up at starting of Path Generation:
DETAIL:
  QUERY
  :commandType 1
  :querySource 0

```

```

:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rttable (
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname partsupp
        :colnames ("ps_partkey" "ps_supkey" "ps_availqty" "ps_supplycost" "p
          s_comment")
      }
    :rtekind 0
    :relid 17191
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname supplier
        :colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
          s_acctbal" "s_comment")
      }
    :rtekind 0
    :relid 17181
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals (
      {OPEXPR
        :opno 1756
        :opfuncid 1720
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 2
            :varattno 6
            :vartype 1700
            :vartypmod 983046
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 2
            :varattnosyn 6
            :location -1
          }
          {CONST
            :consttype 1700
            :consttypmod -1
            :constcollid 0
            :constlen -1
            :constbyval false
            :constisnull false
            :location -1
            :constvalue 6 [ 24 0 0 0 0 -128 ]
          }
        )
      }
    :location -1
  }
  {ALTERNATIVESUBPLAN
    :subplans (
      {SUBPLAN
        :subLinkType 0
        :testexpr <>
        :paramids <>
        :plan_id 1
        :plan_name SubPlan\ 1
        :firstColType 2278
        :firstColTypmod -1
        :firstColCollation 0
        :useHashTable false
        :unknownEqFalse true
        :parallel_safe false
        :setParam <>
        :parParam (i 1)
        :args (
          {VAR
            :varno 2
            :varattno 4
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 2
            :varattnosyn 4
            :location -1
          }
        )
      }
      :startup_cost 0
      :per_call_cost 24.2625
    )
    {SUBPLAN
      :subLinkType 2
      :testexpr
        {OPEXPR
          :opno 96
          :opfuncid 65
          :opresulttype 16
          :opretset false
          :opcollid 0
          :inputcollid 0
          :args (
            {VAR
              :varno 2
              :varattno 4
              :vartype 23
              :vartypmod -1
              :varcollid 0
              :varnullingrels (b)
              :varlevelsup 0
              :varreturningtype 0
              :varnosyn 2
              :varattnosyn 4
              :location -1
            }
            {PARAM
              :paramkind 1
              :paramid 2
            }
          )
        }
      }
    )
  }
)

```

```

        :paramtype 23
        :paramtypmod -1
        :paramcollid 0
        :location -1
    }
    )
    :location -1
    }
    :paramids (i 2)
    :plan_id 2
    :plan_name SubPlan\ 2
    :firstColType 23
    :firstColTypmod -1
    :firstColCollation 0
    :useHashTable true
    :unknownEqFalse true
    :parallel_safe true
    :setParam <>
    :parParam <>
    :args <>
    :startup_cost 24.4875
    :per_call_cost 0.0025
    }
    )
    }
))
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname region
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17191
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17181
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
)
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{PARAM
:paramkind 1
:paramid 0
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location 812
}
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 824
}
)
}
}

```

```

)
:location 822
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 863
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 875
}
)
:location 873
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location 914
}
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 928
}
)
:location 926
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 3
:location 968
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 982
}
)
:location 980
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 4
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 1029
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 1045
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location 1043
}

```

```

    }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {AGGREF
      :aggfnoid 2146
      :aggtype 1700
      :aggcollid 0
      :inputcollid 0
      :aggttranstype 0
      :aggargtypes (o 1700)
      :aggdirectargs <>
      :args
      {TARGETENTRY
        :expr
        {VAR
          :varno 1
          :varattno 4
          :vartype 1700
          :vartypmod 983046
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 4
          :location 599
        }
      }
      :resno 1
      :resname <>
      :ressortgroupref 0
      :resorigtbl 0
      :resorigcol 0
      :resjunk false
    }
  )
  :aggorder <>
  :aggdistinct <>
  :aggfilter <>
  :aggstar false
  :aggvariadic false
  :aggkind n
  :aggpresorted false
  :agglevelsup 0
  :aggsplit 0
  :aggno -1
  :aggttransno -1
  :location 595
}
:resno 1
:resname min
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) (VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 2 :security_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 3 4))) :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 3 4))) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: max_parallel_hazard_walker: Param paramid=0 in_safe_list=0 → reason=outer_reference
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) (VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 2 :security_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 3 4))) :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 3 4))) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = false | reason = PROPAGATE_L_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 1756 :opfncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 2 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1}) (CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0 :plan_id 1 :plan_name SubPlan 1 :firstCollType 2278 :firstCollTypmod -1 :firstCollCollation 0 :useHashable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 1) :args ({VAR :varno 2 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 4 :location -1}) :startup_cost 0 :per_call_cost 24.2625) (SUBPLAN :sublinkType 2 :testexpr {OPEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 2 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 4 :location -1}) (PARAM :paramkind 1 :paramid 2 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :paramIds (i 2) :plan_id 2 :plan_name SubPlan 2 :firstCollType 23 :firstCollTypmod -1 :firstCollCollation 0 :useHashable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 24.4875 :per_call_cost 0.0025)}) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: max_parallel_hazard_walker: SubPlan plan_id=1 parallel_safe=0 → reason=correlated_subplan
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 1756 :opfncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 2 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1}) (CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0 :plan_id 1 :plan_name SubPlan 1 :firstCollType 2278 :firstCollTypmod -1 :firstCollCollation 0 :useHashable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 1) :args ({VAR :varno 2 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 4 :location -1}) :startup_cost 0 :per_call_cost 24.2625) (SUBPLAN :sublinkType 2 :testexpr {OPEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 2 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 4 :location -1}) (PARAM :paramkind 1 :paramid 2 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :paramIds (i 2) :plan_id 2 :plan_name SubPlan 2 :firstCollType 23 :firstCollTypmod -1 :firstCollCollation 0 :useHashable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 24.4875 :per_call_cost 0.0025)}) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = false | reason = PROPAGATE_L_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPAGATE_PARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1}) (VAR :varno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 3 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1}) (VAR :varno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 3 :location -1}) | parallel_safe = true | reason = PROPAGATE_PARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 1054 :opfncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 4 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1}) (CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 2 :security_level 1 :num_base_rels 1 :clause_relds (b 4) :required_relds (b 4) :incompatible_relds (b) :outer_relds (b 4) :right_relds (b) :orclause <> :rinfo_serial 7 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 4 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1} :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 3 4))) :right_em (EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 3 4))) :outer_is_left false :hashjoinoperator

```

```
0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0))
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((VAR :varno 4 :varattno 2 :vartype
1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1) :CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :c
onstbval false :constnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 90 69 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :
has_volatile 2 :security_level 1 :num_base_rels 1 :clause_relds (b 4) :required_relds (b 4) :incompatible_relds (b) :outer_relds (b) :left_relds (b 4) :right_relds (b) :orclause < :rinfo_seria
l 7 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 4 :varattno 2 :vartype 1042 :
vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1) :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 1042 :em_j
domain {JOINDOMAIN :jd_relds (b 1 2 3 4)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbval false :constisnull false :location -1 :
constvalue 10 [ 40 0 0 0 69 85 82 79 90 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4)}} :outer_is_left false :hashjoinoperator
r 8 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0)) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1)
})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 3 :varattno 3 :vartype 23 :va
rtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 3 :location -1) (VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (
b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1)) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof true :
has_volatile 2 :security_level 1 :num_base_rels 2 :clause_relds (b 3 4) :required_relds (b 3 4) :incompatible_relds (b) :outer_relds (b) :left_relds (b 3) :right_relds (b 4) :orclause < :rinfo_seria
l 10 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.00588223529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 3 :varattno 3 :vartype 2
3 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 3 :location -1) :em_relds (b 3) :em_is_const false :em_is_child false :em_datatype 23 :em_jd
omain {JOINDOMAIN :jd_relds (b 1 2 3 4)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :
varnosyn 4 :varattnosyn 1 :location -1) :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4)}} :outer_is_left false :hashjoinoperator 96
:left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96))
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 3 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ({AGGREG :aggfnoid 2146 :aggtype 1700 :aggcollid 0 :inputcollid 0 :aggtranstype 1700 :aggargtypes (o 1700) :aggdirectargs < :args ((TARGETENTRY :expr {VAR :varn
o 1 :varattno 4 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1) :resno 1 :resname < :ressortgroupref 0 :res
orightb 0 :resorigcol 0 :resjunk false}) :aggorder < :aggdistinct < :aggfilter < :aggstar false :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :loc
ation -1))
WARNING: is_parallel_safe: INPUT qual = ({AGGREG :aggfnoid 2146 :aggtype 1700 :aggcollid 0 :inputcollid 0 :aggtranstype 1700 :aggargtypes (o 1700) :aggdirectargs < :args ((TARGETENTRY :expr {VAR :var
no 1 :varattno 4 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1) :resno 1 :resname < :ressortgroupref 0 :res
orightb 0 :resorigcol 0 :resjunk false}) :aggorder < :aggdistinct < :aggfilter < :aggstar false :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :lo
cation -1)) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 4 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :locati
on -1))
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 4 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :locati
on -1}) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: :QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <
:rtable (
  {RANGETBLENTRY
    :alias
      {ALIAS
        :aliasname n
        :colnames <
      }
    :eref
      {ALIAS
        :aliasname n
        :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
      }
    :rtekind 0
    :relid 17158
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <
    :lateral false
    :inFromCl true
    :securityQuals <
  }
  {RANGETBLENTRY
    :alias
      {ALIAS
        :aliasname r
        :colnames <
      }
    :eref
      {ALIAS
        :aliasname r
        :colnames ("r_regionkey" "r_name" "r_comment")
      }
    :rtekind 0
    :relid 17164
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <
    :lateral false
    :inFromCl true
    :securityQuals <
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 17158
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 10)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 17164
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
      {RANGETBLREF
        :rtindex 2
      }
    )
  }
:quals (
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
```

```

:opretset false
:opcollid 0
:inputcollid 0
:args (
  (VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location -1
  )
  (PARAM
    :paramkind 1
    :paramid 3
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
  )
)
:location -1
}
{OEXPR
  :opno 96
  :opfuncid 65
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    (VAR
      :varno 1
      :varattno 3
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 3
      :location -1
    )
    (VAR
      :varno 2
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 1
      :location -1
    )
  )
  :location -1
}
{OEXPR
  :opno 1054
  :opfuncid 1048
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 100
  :args (
    (VAR
      :varno 2
      :varattno 2
      :vartype 1042
      :vartypmod 29
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 2
      :location -1
    )
  )
  {CONST
    :consttype 1042
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
  }
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

```

```

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) {PARAM :paramkind 1 :paramid 3 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 3 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: Param paramid=3 in_safe_list=0 > reason=outer_reference
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) {PARAM :paramkind 1 :paramid 3 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 3 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPAGALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({(VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :co

```

```

nstbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :
leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <
> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :
vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_j
domain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypemod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :cons
tvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :lef
t_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype
1042 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypemod -1 :constcollid 100 :constlen -1 :c
onstbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false
:leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause
<> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042
:vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_
jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypemod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :con
stvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :le
ft_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2: After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:table (
{RANGETBLENTY
:alias
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:reldid 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:reldid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:reldid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:reldid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtdindex 1
}
{RANGETBLREF
:rtdindex 2
}
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypemod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypemod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0

```

```

:varnosyn 2
:varattnosyn 1
:location -1
}
}
:location -1
}
}
{OEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
    :varno 2
    :varattno 2
    :vartype 1042
    :vartypmod 29
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 2
    :location -1
  }
  {CONST
    :consttype 1042
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
  }
)
}
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
}
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = (< | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
{VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
} {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1}) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype
1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :co
nstbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :
leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 2) :right_rels (b) :orclause <
> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :
vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_j
domain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :cons
tvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :lef
t_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype
1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :c
onstbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :
leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 2) :right_rels (b) :orclause
<> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :
vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_
jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :con
stvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :lef
t_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}
) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (
b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}) :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_rels (b 1 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b 2) :orclause <> :rinfo_s
erial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.0058823529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 2
3 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_j
domain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :var
nosyn 2 :varattnosyn 1 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bu
cketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels
(b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}) :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_rels (b 1 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b 2) :orclause <> :rinfo_s
erial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.0058823529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_j
domain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :var
nosyn 2 :varattnosyn 1 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bu
cketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE

```

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WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {
  QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStat <>
  :resultRelation 0
  :hasAggs false
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSublinks true
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity true
  :hasGroupRTE false
  :isReturn false
  :ctelist <>
  :rtable (
    {RANGETBENTRY
    :alias <>
    :eref
    {ALIAS
    :aliasname part
    :colnames ("p_partkey" "p_name" "p_mfg" "p_brand" "p_type" "p_size"
    "p_container" "p_retailprice" "p_comment")
    }
    :rtekind 0
    :relid 17169
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
    {RANGETBENTRY
    :alias <>
    :eref
    {ALIAS
    :aliasname supplier
    :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
    s_acctbal" "s_comment")
    }
    :rtekind 0
    :relid 17181
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals (
      {OPEXPR
      :opno 1756
      :opfuncid 1720
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
        :varno 2
        :varattno 6
        :vartype 1700
        :vartypmod 983046
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 6
        :location -1
        }
        {CONST
        :consttype 1700
        :consttypmod -1
        :constcollid 0
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 6 [ 24 0 0 0 0 -128 ]
        }
      )
    }
    :location -1
    }
  )
  {ALTERNATIVESUBPLAN
  :subplans (
    {SUBPLAN
    :subLinkType 0
    :testexpr <>
    :paramids <>
    :plan_id 4
    :plan_name SubPlan\ 4
    :firstColType 2278
    :firstColTypmod -1
    :firstColCollation 0
    :useHashTable false
    :unknownEqFalse true
    :parallel_safe false
    :setParam <>
    :parParam (i 3)
    :args (
      {VAR
      :varno 2
      :varattno 4
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 4
      :location -1
      }
    )
    :startup_cost 0
    :per_call_cost 24.2625
    }
    {SUBPLAN
    :subLinkType 2
    :testexpr
    {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
      :varno 2
      :varattno 4
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
    }
  )
}

```

```

:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location -1
}
{PARAM
:paramkind 1
:paramid 4
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
}
:location -1
}
:paramids (i 4)
:plan_id 5
:plan_name SubPlan\ 5
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 24.4875
:per_call_cost 0.0025
}
)
}
))
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname partsupp
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
:rtekind 0
:relid 17191
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname region
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 5
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17169
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17181
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17191
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
):jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF

```

```

:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
{RANGETBLREF
:rtindex 5
}
}
:qual (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 283
}
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 295
}
)
:location 293
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 318
}
{VAR
:varno 3
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location 330
}
)
:location 328
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 353
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 362
:constvalue 4 [ 15 0 0 0 0 0 0 0 ]
}
)
:location 360
}
{OEXPR
:opno 1209
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{RELABELTYPE
:arg
{VAR
:varno 1
:varattno 5
:vartype 1043
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 377
}
:resulttype 25
:resultypmod -1
:resultcollid 100
:relabelformat 2
:location -1
}
{CONST

```

```

:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 389
:constvalue 10 [ 40 0 0 0 37 66 82 65 83 83 ]
}
)
:location 384
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location 410
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 424
}
}
:location 422
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location 448
}
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location 462
}
}
:location 460
}
{OEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 5
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 2
:location 493
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 509
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
}
:location 507
}
{OEXPR
:opno 1752
:opfuncid 1718
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 4
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 4
:location 530
}
{SUBPLAN
:sublinkType 4
:testexpr <>
:paramids <>
:plan_id 3
:plan_name SubPlan\ 3
:firstColType 1700

```

```

:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse false
:parallel_safe false
:setParam <>
:parParam (i 0)
:args (
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 812
  }
)
:startup_cost 0
:per_call_cost 135379.985000000004
}
:location 544
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 2
      :varattno 6
      :vartype 1700
      :vartypmod 983046
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 6
      :location 51
    }
    :resno 1
    :resname s_acctbal
    :ressortgroupref 1
    :resorigtbl 17181
    :resorigcol 6
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 2
      :varattno 2
      :vartype 1042
      :vartypmod 29
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 2
      :location 70
    }
    :resno 2
    :resname s_name
    :ressortgroupref 3
    :resorigtbl 17181
    :resorigcol 2
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 4
      :varattno 2
      :vartype 1042
      :vartypmod 29
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 4
      :varattnosyn 2
      :location 86
    }
    :resno 3
    :resname n_name
    :ressortgroupref 2
    :resorigtbl 17158
    :resorigcol 2
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location 102
    }
    :resno 4
    :resname p_partkey
    :ressortgroupref 4
    :resorigtbl 17169
    :resorigcol 1
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 3
      :vartype 1042
      :vartypmod 29
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 3
      :location 121
    }
    :resno 5
    :resname p_mfgr
    :ressortgroupref 0
    :resorigtbl 17169
    :resorigcol 3
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 2

```

```

:varattno 3
:vartype 1043
:vartypmod 44
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location 137
}
:resno 6
:resname s_address
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 5
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 5
:location 156
}
:resno 7
:resname s_phone
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 7
:vartype 1043
:vartypmod 105
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 7
:location 173
}
:resno 8
:resname s_comment
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 7
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tltSortGroupRef 1
:eqop 1752
:sortop 1756
:reverse_sort true
:nulls_first true
:hashable true
}
{SORTGROUPCLAUSE
:tltSortGroupRef 2
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
{SORTGROUPCLAUSE
:tltSortGroupRef 3
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
{SORTGROUPCLAUSE
:tltSortGroupRef 4
:eqop 96
:sortop 97
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount
{CONST
:consttype 20
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 100 0 0 0 0 0 0 0 ]
}
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

```

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1209 :opfuncid 850 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({RELABELTYPE :arg {VAR :varno 1 :var
attno 5 :vartype 1043 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1} :resulttype 25 :resulttypmod -1 :resultcollid 100 :rela
belformat 2 :location -1} {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 37 66 82 65 83 83 ]}) :location -
1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (
b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 6 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeopfami
lies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RES
TRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :v
arlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [
15 0 0 0 0 0 0 0 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 2 :security_level 1 :num_base_rels 1 :clause
_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 5 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select
-1 :outer_select -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 6 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0
:varnosyn 1 :varattnosyn 6 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4 5)}} :right_em {EQUIVALENCEMEMBER :em_expr
{CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ 15 0 0 0 0 0 0 0 ]} :em_relds (b) :em_is_const true :em_is_child false :
em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4 5)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator
0 :right_hashoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1209 :opfuncid 850 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({RELABELTYPE :arg {VAR :varno 1 :va
rattno 5 :vartype 1043 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1} :resulttype 25 :resulttypmod -1 :resultcollid 100 :rel
abelformat 2 :location -1} {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 37 66 82 65 83 83 ]}) :location
-1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds

```


WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 2 :varattno 6 :vartype 1700 :vartypmod 983846 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :locat ion -1) {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {VAR :varno 4 :varattno 2 :va rtype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 : varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 3 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :va rreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 3 :vartype 1043 :vartypmod 44 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattn osyn 3 :location -1} {VAR :varno 2 :varattno 5 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 5 :location -1} {VAR :varno 2 :v arattno 7 :vartype 1043 :vartypmod 105 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 7 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 2 :varattno 6 :vartype 1700 :vartypmod 983846 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :locati on -1) {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {VAR :varno 4 :varattno 2 :var tpe 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :v arnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 3 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :va rreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 3 :vartype 1043 :vartypmod 44 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattno syn 3 :location -1} {VAR :varno 2 :varattno 5 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 5 :location -1} {VAR :varno 2 :va rattno 7 :vartype 1043 :vartypmod 105 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 7 :location -1})

WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 2 :varattno 6 :vartype 1700 :vartypmod 983846 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :locat ion -1) {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {VAR :varno 4 :varattno 2 :var tpe 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :v arnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 3 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :va rreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 3 :vartype 1043 :vartypmod 44 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattn osyn 3 :location -1} {VAR :varno 2 :varattno 5 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 5 :location -1} {VAR :varno 2 :v arattno 7 :vartype 1043 :vartypmod 105 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 7 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE

QUERY PLAN

Limit (cost=31682955.10..31682955.11 rows=1 width=271) (actual time=24232.882..24232.909 rows=100.00 loops=1)

Output: supplier.s_acctbal, supplier.s_name, nation.n_name, part.p_partkey, part.p_mfgr, supplier.s_address, supplier.s_phone, supplier.s_comment

Buffers: shared hit=6432267, temp read=863 written=863

→ Sort (cost=31682955.10..31682955.11 rows=1 width=271) (actual time=24232.881..24232.901 rows=100.00 loops=1)

Output: supplier.s_acctbal, supplier.s_name, nation.n_name, part.p_partkey, part.p_mfgr, supplier.s_address, supplier.s_phone, supplier.s_comment

Sort Key: supplier.s_acctbal DESC, nation.n_name, supplier.s_name, part.p_partkey

Sort Method: quicksort Memory: 77kB

Buffers: shared hit=6432267, temp read=863 written=863

→ Hash Join (cost=135835.08..31682955.09 rows=1 width=271) (actual time=400.720..24232.486 rows=190.00 loops=1)

Output: supplier.s_acctbal, supplier.s_name, nation.n_name, part.p_partkey, part.p_mfgr, supplier.s_address, supplier.s_phone, supplier.s_comment

Hash Cond: (((SubPlan 3) = partsupp.ps_supplycost) AND (part.p_partkey = partsupp.ps_partkey))

Buffers: shared hit=6432258, temp read=863 written=863

→ Seq Scan on public.part (cost=0.00..3580.00 rows=466 width=30) (actual time=0.043..15.224 rows=362.00 loops=1)

Output: part.p_partkey, part.p_name, part.p_mfgr, part.p_brand, part.p_type, part.p_size, part.p_container, part.p_retailprice, part.p_comment

Filter: ((part.p_size = 15) AND ((part.p_type)::text ~> '%BRASS':text))

Rows Removed by Filter: 99638

Buffers: shared hit=2080

→ Hash (cost=135826.26..135826.26 rows=588 width=251) (actual time=128.065..128.070 rows=70960.00 loops=1)

Output: supplier.s_acctbal, supplier.s_name, supplier.s_address, supplier.s_phone, supplier.s_comment, partsupp.ps_partkey, partsupp.ps_supplycost, nation.n_name

Buckets: 65536 (originally 1024) Batches: 2 (originally 1) Memory Usage: 7763kB

Buffers: shared hit=9356, temp written=858

→ Hash Join (cost=121528.66..135826.26 rows=588 width=251) (actual time=6.223..88.852 rows=70960.00 loops=1)

Output: supplier.s_acctbal, supplier.s_name, supplier.s_address, supplier.s_phone, supplier.s_comment, partsupp.ps_partkey, partsupp.ps_supplycost, nation.n_name

Hash Cond: (partsupp.ps_supkey = supplier.s_supkey)

Buffers: shared hit=9356

→ Seq Scan on public.partsupp (cost=0.00..12792.00 rows=400000 width=14) (actual time=0.011..26.774 rows=400000.00 loops=1)

Output: partsupp.ps_partkey, partsupp.ps_supkey, partsupp.ps_availqty, partsupp.ps_supplycost, partsupp.ps_comment

Buffers: shared hit=8792

→ Hash (cost=121528.58..121528.58 rows=7 width=245) (actual time=6.203..6.207 rows=887.00 loops=1)

Output: supplier.s_acctbal, supplier.s_name, supplier.s_address, supplier.s_phone, supplier.s_comment, supplier.s_supkey, nation.n_name

Buckets: 1024 Batches: 1 Memory Usage: 184kB

Buffers: shared hit=564

→ Nested Loop (cost=0.00..121528.58 rows=7 width=245) (actual time=0.071..5.910 rows=887.00 loops=1)

Output: supplier.s_acctbal, supplier.s_name, supplier.s_address, supplier.s_phone, supplier.s_comment, supplier.s_supkey, nation.n_name

Join Filter: (supplier.s_nationkey = nation.n_nationkey)

Rows Removed by Join Filter: 3548

Buffers: shared hit=564

→ Nested Loop (cost=0.00..25.95 rows=1 width=108) (actual time=0.015..0.019 rows=5.00 loops=1)

Output: nation.n_name, nation.n_nationkey

Join Filter: (nation.n_regionkey = region.r_regionkey)

Rows Removed by Join Filter: 20

Buffers: shared hit=2

→ Seq Scan on public.region (cost=0.00..12.12 rows=1 width=4) (actual time=0.006..0.007 rows=1.00 loops=1)

Output: region.r_regionkey, region.r_name, region.r_comment

Filter: ((region.r_name = 'EUROPE')::bpchar)

Rows Removed by Filter: 4

Buffers: shared hit=1

→ Seq Scan on public.nation (cost=0.00..11.70 rows=170 width=112) (actual time=0.006..0.007 rows=25.00 loops=1)

Output: nation.n_nationkey, nation.n_name, nation.n_regionkey, nation.n_comment

Buffers: shared hit=1

→ Seq Scan on public.supplier (cost=0.00..121487.00 rows=1250 width=145) (actual time=0.009..1.124 rows=887.00 loops=5)

Output: supplier.s_acctbal, supplier.s_name, supplier.s_address, supplier.s_phone, supplier.s_comment, supplier.s_supkey, nation.n_name

Filter: (((supplier.s_acctbal > '0')::numeric) AND (ANY (supplier.s_nationkey = (hashed SubPlan 5).col1)))

Rows Removed by Filter: 4113

Buffers: shared hit=562

SubPlan 5

→ Hash Join (cost=12.14..24.48 rows=1 width=4) (actual time=0.018..0.024 rows=5.00 loops=1)

Output: n_1.n_nationkey

Hash Cond: (n_1.n_n_regionkey = r_1.r_regionkey)

Buffers: shared hit=2

→ Seq Scan on public.nation n_1 (cost=0.00..11.70 rows=170 width=8) (actual time=0.002..0.004 rows=25.00 loops=1)

Output: n_1.n_nationkey, n_1.n_name, n_1.n_regionkey, n_1.n_comment

Buffers: shared hit=1

→ Hash (cost=12.12..12.12 rows=1 width=4) (actual time=0.009..0.009 rows=1.00 loops=1)

Output: r_1.r_regionkey

Filter: (r_1.r_name = 'EUROPE')::bpchar)

Buckets: 1024 Batches: 1 Memory Usage: 9kB

Buffers: shared hit=1

→ Seq Scan on public.region r_1 (cost=0.00..12.12 rows=1 width=4) (actual time=0.003..0.003 rows=1.00 loops=1)

Output: r_1.r_regionkey

Filter: (r_1.r_name = 'EUROPE')::bpchar)

Rows Removed by Filter: 4

Buffers: shared hit=1

SubPlan 3

→ Aggregate (cost=135379.98..135379.99 rows=1 width=32) (actual time=33.395..33.396 rows=1.00 loops=721)

Output: min(partsupp_1.ps_supplycost)

Buffers: shared hit=6420822

→ Nested Loop (cost=0.00..135379.97 rows=1 width=6) (actual time=32.951..33.394 rows=0.72 loops=721)

Output: partsupp_1.ps_supplycost

Join Filter: (nation_1.r_regionkey = region_1.r_regionkey)

Buffers: shared hit=6420822

→ Nested Loop (cost=0.00..135367.84 rows=1 width=10) (actual time=32.948..33.391 rows=0.72 loops=721)

Output: partsupp_1.ps_supplycost, nation_1.n_nationkey

Join Filter: (supplier_1.s_nationkey = nation_1.n_nationkey)

Rows Removed by Join Filter: 17

Buffers: shared hit=6420804

→ Nested Loop (cost=0.00..135354.01 rows=1 width=10) (actual time=32.939..33.380 rows=0.72 loops=721)

Output: partsupp_1.ps_supplycost, supplier_1.s_nationkey

Join Filter: (partsupp_1.ps_supkey = supplier_1.s_supkey)

Rows Removed by Join Filter: 3547

Buffers: shared hit=6419786

→ Seq Scan on public.supplier supplier_1 (cost=0.00..121487.00 rows=1250 width=8) (actual time=0.002..1.160 rows=887.00 loops=721)

Output: supplier_1.s_supkey, supplier_1.s_name, supplier_1.s_address, supplier_1.s_nationkey, supplier_1.s_phone, supplier_1.s_acctbal, supplier_1.s_comme

nt

Filter: (((supplier_1.s_acctbal > '0')::numeric) AND (ANY (supplier_1.s_nationkey = (hashed SubPlan 2).col1)))

Rows Removed by Filter: 4113

Buffers: shared hit=80754

SubPlan 2

→ Hash Join (cost=12.14..24.48 rows=1 width=4) (actual time=0.032..0.037 rows=5.00 loops=1)

Output: n.n_nationkey

Hash Cond: (n.n_regionkey = r.r_regionkey)

Buffers: shared hit=2

→ Seq Scan on public.nation n (cost=0.00..11.70 rows=170 width=8) (actual time=0.005..0.007 rows=25.00 loops=1)

Output: n.n_nationkey, n.n_name, n.n_regionkey, n.n_comment

Buffers: shared hit=1

→ Hash (cost=12.12..12.12 rows=1 width=4) (actual time=0.014..0.014 rows=1.00 loops=1)

Output: r.r_regionkey

Buckets: 1024 Batches: 1 Memory Usage: 9kB

Buffers: shared hit=1

→ Seq Scan on public.region r (cost=0.00..12.12 rows=1 width=4) (actual time=0.011..0.012 rows=1.00 loops=1)

Output: r.r_regionkey

Filter: (r.r_name = 'EUROPE')::bpchar)

Rows Removed by Filter: 4

Buffers: shared hit=1

→ Materialize (cost=0.00..13792.02 rows=4 width=10) (actual time=0.018..0.036 rows=4.00 loops=639527)

Output: partsupp_1.ps_supplycost, partsupp_1.ps_supkey

Storage: Memory Maximum Storage: 17kB

Buffers: shared hit=6339032

→ Seq Scan on public.partsupp partsupp_1 (cost=0.00..13792.00 rows=4 width=10) (actual time=15.864..31.775 rows=4.00 loops=721)

Output: partsupp_1.ps_supplycost, partsupp_1.ps_supkey

Filter: (part.p_partkey = partsupp_1.ps_partkey)

Rows Removed by Filter: 399996

Buffers: shared hit=6339032

→ Seq Scan on public.nation nation_1 (cost=0.00..11.70 rows=170 width=8) (actual time=0.001..0.002 rows=25.00 loops=518)